you'll learn why testing programs with dependencies is a challenge, how you can isolate those dependencies, and how to deal with them by using *test doubles*.

10.3.1. Why dependencies make testing difficult

Remember the task at hand: rewrite the tracking application using a test-driven approach. One part of the application is a dashboard that Agtron uses to see what's happening on his web shop: how many visitors there are, what they're looking at, and things like that. Imagine you're writing a test for part of the dashboard that extracts some data obtained via an HTTP service that returns JSON:

```
fact "data is parsed correctly", ->
  extractData 'support', (res) ->
    assert.equals res.status, 'online'
```

A simplified example of data from the HTTP service contains just one property and value:

```
{ status: 'online' }
```

You implement `extractData` to take a callback and pass it as the callback to `http.get`:

```
http = require 'http'

extractData = (topic, callback) ->
  options =
    host: 'www.agtronsemporium.com'
    port: 80
    path: "/service/#{topic}"

  http.get(options, (res) ->
    callback res.something
  ).on 'error', (e) ->
    console.log e
```

This works well until the day you crash the website. When www.agtronsemporium.com crashes, you're no longer able to get data from